

Section 1.1

1.1.8

Write the number in the form $a + bi$:

$$\frac{(8 + 2i) - (1 - i)}{(2 + i)^2}$$

Solution. Rearranging:

$$\begin{aligned}\frac{(8 + 2i) - (1 - i)}{(2 + i)^2} &= \frac{(8 + 2i) - (1 - i)}{3 + 4i} \\ &= \frac{(8 + 2i) - (1 - i)}{3 + 4i} \cdot \frac{3 - 4i}{3 - 4i} \\ &= \frac{33 - 19i}{25} \\ &= \frac{33}{25} - \frac{19}{25}i\end{aligned}$$

□

1.1.10

Write the number in the form $a + bi$:

$$\left[\frac{2 + i}{6i - (1 - 2i)} \right]^2$$

Solution.

$$\begin{aligned}\left(\frac{2 + i}{6i - (1 - 2i)} \right)^2 &= \frac{(2 + i)^2}{(-1 + 8i)^2} \\ &= \frac{4 + 4i - 1}{1 - 16i - 64} \\ &= \frac{3 + 4i}{-63 - 16i} \\ &= \frac{(3 + 4i)(-63 + 16i)}{(-63 - 16i)(-63 + 16i)} \\ &= \frac{-189 + 48i - 252i - 64}{63^2 + 16^2} \\ &= \frac{-253 - 204i}{63^2 + 16^2} \\ &= \frac{-253}{63^2 + 16^2} + \frac{-204i}{63^2 + 16^2}\end{aligned}$$

□

1.1.14

Show that $\operatorname{Re}(iz) = -\operatorname{Im} z$ for every complex number z .

Proof. Write $z = x + yi$ where $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$. Then $iz = xi - y$ and $\operatorname{Re}(iz) = -y = -\operatorname{Im}(z)$. $\square$

1.1.15

Let k be an integer. Show that

$$i^{4k} = 1, \quad i^{4k+1} = i, \quad i^{4k+2} = -1, \quad i^{4k+3} = -i.$$

Proof. **1.**

$$i^{4k} = (i^2)^{2k} = ((-1)^2)^k = 1^k$$

Because $1^k = 1 \quad \forall k \in \mathbb{Z}$.

2.

$$\begin{aligned} i^{4k+1} &= i^{4k} \cdot i \\ &= 1 \cdot i && \text{we can substitute the result of 1.1.15 part 1} \\ &= i \end{aligned}$$

3.

$$\begin{aligned} i^{4k+2} &= i^{4k} \cdot i^2 \\ &= 1 \cdot i^2 && \text{using the result from 1.1.15 part 1} \\ &= i^2 \\ &= -1 \end{aligned}$$

4.

$$\begin{aligned} i^{4k+3} &= i^{4k} \cdot i^3 \\ &= 1 \cdot i^3 && \text{using the result from 1.1.15 part 1} \\ &= i^3 \\ &= i^2 \cdot i \\ &= -1 \cdot i \\ &= -i \end{aligned}$$

$\square$

1.1.20

Solve each of the following equations for z .

1.1.20 (c)

$$(2 - i)z + 8z^2 = 0$$

Solution.

$$\begin{aligned}(2 - i)z + 8z^2 &= 0 \\ z((2 - i) + 8z) &= 0\end{aligned}$$

Thus:

$$z = 0 \text{ or } z = \frac{-2 + i}{8}$$

□

1.1.22

Find all solutions to the equation $z^4 - 16 = 0$.

Solution.

$$\begin{aligned}z^4 - 16 &= 0 \\ (z^2 - 4)(z^2 + 4) &= 0\end{aligned}$$

Thus we solve each root independently:

$$\begin{aligned}(z^2 - 4) = 0 &\implies z = \pm 2 \\ (z^2 + 4) = 0 &\implies z = \pm 2i\end{aligned}$$

Thus all solutions are: $z = \{2, -2, 2i, -2i\}$.

□

1.1.24

Let z be a complex number such that $\text{Im } z > 0$. Prove that $\text{Im}(1/z) < 0$.

Proof. Let $z = x + yi$ with $\text{Im}(z) > 0$, so $y > 0$. Thus:

$$\begin{aligned}\frac{1}{z} &= \frac{1}{x + yi} \\ &= \frac{x - yi}{x^2 + y^2} && \text{multiplying by the conjugate} \\ &= \frac{x}{x^2 + y^2} + \frac{-yi}{x^2 + y^2} && \text{separating the real and imaginary components}\end{aligned}$$

Thus taking the imaginary component:

$$\text{Im}\left(\frac{1}{z}\right) = \frac{-y}{x^2 + y^2}$$

The denominator is positive because any a^2 for $a \in \mathbb{R}$ is positive and sums of positives are also positive. The numerator is negative because $y > 0$ so $-y < 0$. Finally we know the whole fraction is negative: $u, v \in \mathbb{R}, u < 0, v > 0 \implies \frac{u}{v} < 0$. Thus:

$$\operatorname{Im} \left(\frac{1}{z} \right) < 0$$

□

Section 1.2

1.2.7

Describe the set of points z in the complex plane that satisfies each of the following.

1.2.7 (d)

$$|z - 1| = |z + i|$$

Solution.

$$\begin{aligned} |z - 1| &= |z + i| \\ \sqrt{(x - 1)^2 + y^2} &= \sqrt{x^2 + (y + 1)^2} && \text{substituting } z = x + yi \\ (x - 1)^2 + y^2 &= x^2 + (y + 1)^2 && \text{squaring both sides} \\ x^2 - 2x + 1 + y^2 &= x^2 + y^2 + 2y + 1 \\ -2x &= 2y && \text{canceling terms} \\ y &= -x \end{aligned}$$

Thus the set satisfying $|z - 1| = |z + i|$ is $\{x + yi : y = -x, x, y \in \mathbb{R}\}$, the perpendicular bisector of 1 and $-i$. □

1.2.7 (e)

$$|z| = \operatorname{Re} z + 2$$

Solution.

$$\begin{aligned} |z| &= \operatorname{Re}(z) + 2 \\ \sqrt{x^2 + y^2} &= x + 2 && \text{substituting } z = x + yi \text{ such that } \operatorname{Re}(z) = x \\ x^2 + y^2 &= x^2 + 4x + 4 && \text{squaring both sides} \\ y^2 &= 4x + 4 \\ y &= \pm\sqrt{4x + 4} \end{aligned}$$

Thus the set satisfying $|z| = \operatorname{Re}(z) + 2$ is $\{x + yi : y = \pm\sqrt{4x + 4}, x, y \in \mathbb{R}\}$. □

1.2.7 (f)

$$|z - 1| + |z + 1| = 7$$

Solution. We begin with:

$$|z - 1| + |z + 1| = 7$$

Substituting in the following for the absolutes and naming them A and B respectively:

$$A = |z - 1| = \sqrt{(x - 1)^2 + y^2}$$

$$B = |z + 1| = \sqrt{(x + 1)^2 + y^2}$$

We get:

$$A = 7 - B$$

$$A^2 = (7 - B)^2$$

we can square both sides

and substituting the expression for A from the definition of the absolute:

$$(x - 1)^2 + y^2 = (7 - B)^2$$

$$(x - 1)^2 + y^2 = 49 - 14B + B^2$$

substituting the expression for B from the definition of the absolute:

$$= 49 - 14B + (x + 1)^2 + y^2$$

$$(x - 1)^2 + y^2 = 49 - 14B + (x + 1)^2 + y^2$$

and canceling terms

$$x^2 - 2x + 1 + y^2 = 49 - 14B + x^2 + 2x + 1 + y^2$$

$$14B = 4x + 49$$

simplifying

finally substituting in B for its definition from the absolute and simplifying further:

$$49 = 14\sqrt{(x + 1)^2 + y^2} - 4x$$

$$49^2 = \left(14\sqrt{(x + 1)^2 + y^2} - 4x\right)^2$$

$$49^2 + 392x + 16x^2 = 14^2(x^2 + 2x + 1 + y^2)$$

$$0 = 180x^2 + 196 - 49^2 + 169y^2$$

$$2205 = 180x^2 + 196y^2$$

Thus the set satisfying $|z - 1| + |z + 1| = 7$ is:

$$\{x + yi : 2205 = 180x^2 + 196y^2, x, y \in \mathbb{R}\}$$

□

1.2.16

Prove that if $|z| = 1$ ($z \neq 1$), then

$$\operatorname{Re} \left[\frac{1}{1-z} \right] = \frac{1}{2}.$$

Proof. Rearranging $\frac{1}{1-z}$ to separate the real and imaginary components, we get:

$$\begin{aligned} \frac{1}{1-z} &= \frac{1}{1-(x+yi)} \\ &= \frac{1}{(1-x)-yi} \cdot \frac{(1-x)+yi}{(1-x)+yi} \end{aligned}$$

which is well defined because $z \neq 1 \implies$ the denominator $(1-x)+yi \neq 0$

$$\begin{aligned} &= \frac{(1-x)+yi}{(1-x)^2+y^2} \\ &= \frac{1-x}{(1-x)^2+y^2} + \frac{yi}{(1-x)^2+y^2} \end{aligned}$$

Thus taking just the real component:

$$\operatorname{Re} \left(\frac{1}{1-z} \right) = \frac{1-x}{(1-x)^2+y^2}$$

Now from $|z| = 1$ we can expand the absolute and rearrange algebraically:

$$\begin{aligned} 1 &= |z| \\ &= \sqrt{x^2+y^2} \\ &= x^2+y^2 \\ 2-2x &= x^2-2x+1+y^2 \\ &= (x-1)^2+y^2 \\ 1 &= \frac{2-2x}{(x-1)^2+y^2} \end{aligned}$$

denominator is nonzero because $z \neq 1 \implies (x-1)^2+y^2 \neq 0$

$$\frac{1}{2} = \frac{1-x}{(x-1)^2+y^2}$$

Which is exactly $\operatorname{Re} \left(\frac{1}{1-z} \right)$. Thus:

$$\operatorname{Re} \left(\frac{1}{1-z} \right) = \frac{1}{2}$$

□

Section 1.3

1.3.7

Find the argument of each of the following complex numbers and write each in polar form.

1.3.7 (d)

$$-2\sqrt{3} - 2i$$

Solution. $z = -2\sqrt{3} - 2i$. Find the argument angle θ and account for the fact that $\arctan$'s range is always within the 1st or 4th quadrants:

$$\begin{aligned}\tan \theta &= \frac{-2}{-2\sqrt{3}} \implies \theta = \frac{\pi}{6} \\ \theta + \pi &= \frac{7\pi}{6} \text{ or equivalently } -\frac{5\pi}{6}\end{aligned}$$

The absolute is:

$$|-2\sqrt{3} - 2i| = \sqrt{(-2\sqrt{3})^2 + (-2)^2} = 4$$

Thus we write:

$$z = 4 \left(\cos \left(-\frac{5\pi}{6} \right) + i \sin \left(-\frac{5\pi}{6} \right) \right)$$

□

1.3.7 (g)

$$\frac{-1 + \sqrt{3}i}{2 + 2i}$$

Solution. We begin by separating the real and imaginary components of $z = \frac{-1 + \sqrt{3}i}{2 + 2i}$ by multiplying the numerator and denominator by the conjugate of the denominator:

$$z = \frac{(-2 + 2\sqrt{3}) + i(2 + 2\sqrt{3})}{8}$$

Then taking the inverse tangent:

$$\theta = \arctan \left(\frac{2 + 2\sqrt{3}}{-2 + 2\sqrt{3}} \right) = \frac{5}{12}\pi \quad \text{which is already in the 1st quadrant}$$

And computing the absolute:

$$r = |z| = \frac{\sqrt{2}}{2}$$

Finally:

$$z = \frac{\sqrt{2}}{2} \left(\cos \left(\frac{5}{12}\pi \right) + i \sin \left(\frac{5}{12}\pi \right) \right)$$

□

1.3.10

Show the following:

1.3.10 (a)

$$\arg(z_1 z_2 z_3) = \arg z_1 + \arg z_2 + \arg z_3$$

Proof. We know that for two complex numbers $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$, $z_2 = r_2(\cos \theta_2 + i \sin \theta_2) \in \mathbb{C}$:

$$z_1 \cdot z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Thus, we can apply this to z_1, z_2, z_3 by writing $z_1 \cdot z_2$ as z_a :

$$z_a = r_a(\cos(\theta_a) + i \sin(\theta_a)) \quad \text{with } r_a = r_1 \cdot r_2, \quad \theta_a = \theta_1 + \theta_2$$

Then:

$$\begin{aligned} z_1 z_2 z_3 &= z_a \cdot z_3 = r_a r_3 [\cos(\theta_a + \theta_3) + i \sin(\theta_a + \theta_3)] \\ &= r_1 r_2 r_3 [\cos(\theta_1 + \theta_2 + \theta_3) + i \sin(\theta_1 + \theta_2 + \theta_3)] \end{aligned}$$

Taking the argument from the sin and cos terms:

$$\arg(z_1 z_2 z_3) = \theta_1 + \theta_2 + \theta_3$$

□

1.3.10 (b)

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

Proof. We can write the argument of $z_2 = x + iy \in \mathbb{C}$ as $\arg(z_2) = \arctan\left(\frac{y}{x}\right) = \theta_2$.

Taking the conjugate of z_2 , then solving for its argument:

$$\overline{z_2} = x + i(-y)$$

Which can be written as $r_2(\cos \theta + i \sin \theta)$ for some angle θ . Solving for θ by setting the real and imaginary parts equal:

$$\begin{aligned} x &= r_2 \cos \phi \\ y &= r_2 \sin \phi \\ \frac{-y}{x} &= \tan \phi \end{aligned}$$

This leaves:

$$\begin{aligned} \theta &= \arctan\left(-\frac{y}{x}\right) \implies -\phi = \arctan\left(\frac{y}{x}\right) = \theta_2 \\ \therefore z &= r_2(\cos(-\theta_2) + i \sin(-\theta_2)) \end{aligned}$$

Now we can apply the known identity $z_1 \cdot z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$ such that:

$$z_1 \overline{z_2} = r_1 r_2 [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

and we can read off the argument:

$$\arg(z_1 \overline{z_2}) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$$

□

1.3.12

Find the following.

1.3.12 (b)

$$\text{Arg}(-\pi)$$

Solution. $-\pi$ is purely real valued and negative so its argument is $\arg_0 = \pi$.

□

1.3.12 (c)

$$\text{Arg}(10i)$$

Solution. $10i$ is purely imaginary so its argument is $\arg_0 = \frac{\pi}{2}$.

□

Additional Problem

$$\text{Re}(zw) = \text{Re}(z) \cdot \text{Re}(w)$$

Proof. It is not true that $\text{Re}(zw) = \text{Re}(z) \text{Re}(w)$.

Counterexample: $z = 1 + i$, $w = 1 + i$.

Plugging into the LHS:

$$\text{Re}(zw) = \text{Re}((1 + i)(1 + i)) = \text{Re}(1 + 2i - 1) = \text{Re}(2i) = 0$$

Plugging into the RHS:

$$\text{Re}(1 + i) \cdot \text{Re}(1 + i) = 1$$

Thus: $\text{Re}(zw) \neq \text{Re}(z) \text{Re}(w)$.

□